SYSTEMS MANAGEMENT



OCP AMC v 2.0
(Auxiliary Management Controller) (fka SMC)

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Problem

- Every PCIe, CXL or other component has its own Management Interface style
  - BMC must have special code for every vendor & every product from that vendor
- MCTP & PLDM helps
  - MCTP has mappings for I2C/I3C, PCIe, CXL, Serial and USB
  - MCTP/PLDM has tons of useful things: SPDM, FW Update, RDE, FRU, etc.
- But there are too many options in these specs
  - And no single spec exists that combines requirements (think Profile)
  - And there is no open-source project for those that use an interposer
- Vendors have projects & specs:
  - Meta: OpenBIC for Servers, Google: Mini-BMC for Storage, Samsung: DMC for Storage, HPE and other systems integrators have their own specs too

AMC Specification

Define the Software/Firmware API requirements for managing auxiliary devices

- "Profile" for MCTP/PLDM & SPDM Specs

Complementary to a BMC

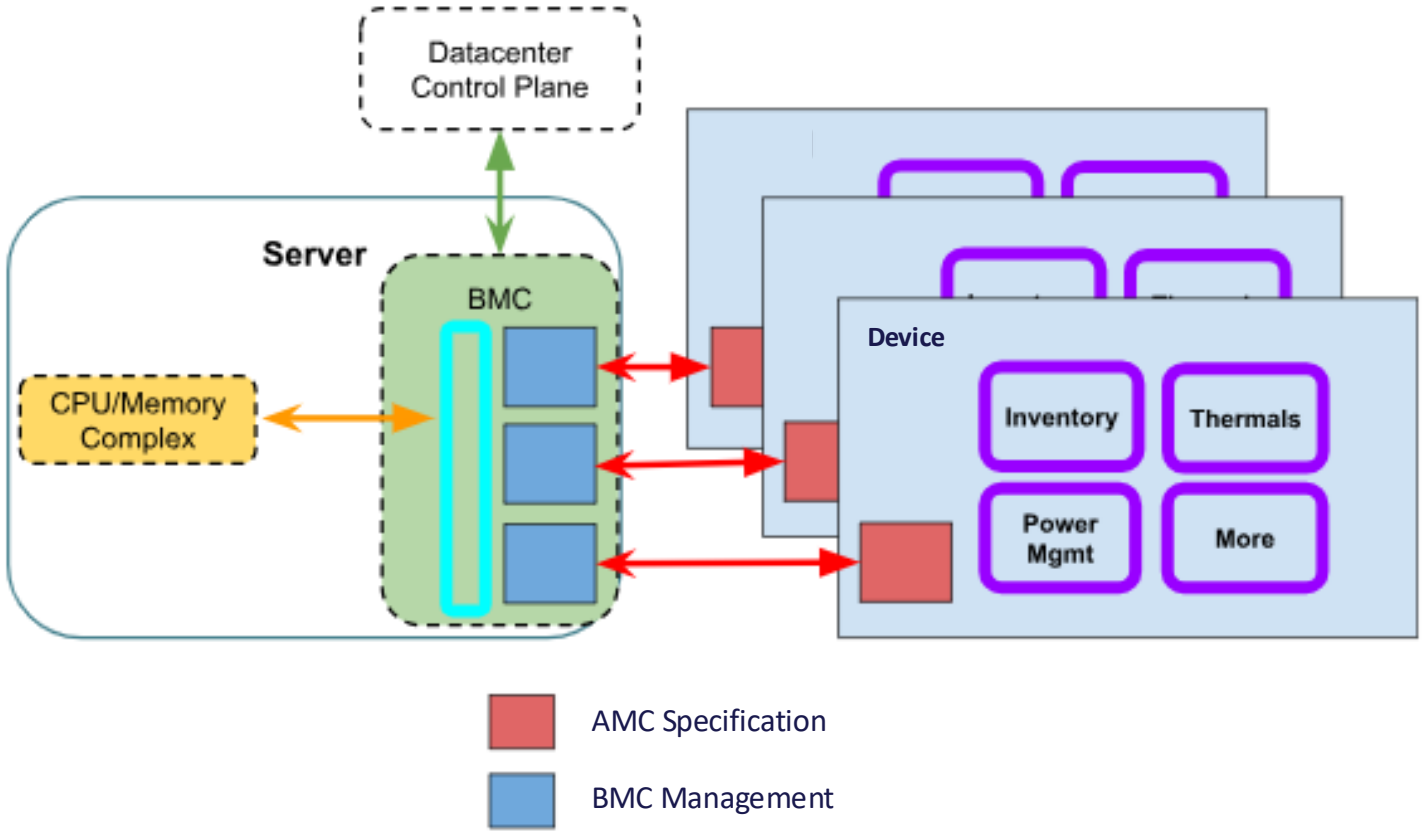
- Can be added interposer or existing HW/FW space. Existing HW/FW space can provide savings

Target devices are auxiliary devices to the main board

- Examples: Accelerators, Storage Controllers, Drives, NICs, DPUs, SoCs, etc.

AMCs are **not** a replacement for BMCs, they are complementary

- Instead they feed information to the BMC
- Can either be in existing FW footprint or an interposer chip



What's in the AMC Spec

- AMC-to-BMC protocol
  - Proscriptive Requirements for PLDM, MCTP, I2C, RDE, PCI-VDM
    - Spec has the commands and requirements for each of these specifications
    - Redfish Device Enablement (RDE) requirements
      - Validated with Profiles from the Profiles Workstream using open-source tools
      - HPE Options support from many vendors (Broadcom, Intel, Marvel, Microchip, Nvidia/Mellanox)
    - Timing parameters for commands are included
- Conformance
    - Can be validated using the Redfish Interop Validator
    - Thermal Properties, Fan Speed, FRU ID & Fields (Manufacturer)
    - Software Inventory, Thermal Setpoint, Power Management
    - Security (SPDM) Requirements

What's missing?

- The Redfish interop profiles to go along with the spec
  - OCP has an interop profile subproject under Management project
    - https://github.com/opencomputeproject/HWMgmt-OCP-Profiles
    - Ethernet NIC profile is an example of one such profile (final)
- Conformance Tests for functional interoperability
  - DMTF's Redfish Forum has an open-source conformance checker that is based on Redfish Interop Profiles
- Code
  - Open-Source project to assist devices (auxiliary management controllers)
  - Google open source at https://github.com/google/smc-common

SMC-Common (library for AMC firmware)

- Open-source foundational reference implementation framework from Google
  - AMC Project used to be called SMC.
  - https://github.com/google/smc-common
- Zephyr RTOS-based library for building AMC/SMC firmware
- Provides reusable infrastructure:
  - RDE server
  - PLDM base support
  - MCTP transport support
  - Sensor framework
  - FRU support
  - Logging
  - GPIO and watchdog services
  - Device drivers for common management devices

AMC 2.0 – Why the Update

- Key requirements were added or further refined
- Syncing requirements with other key industry and OCP specifications involving device management.
- Address missing features not included in 1.0
  - Making these required forced the version to 2.0

AMC 2.0 Major Changes

- Security / SPDM requirements fully defined
  - Response code, certificate and firmware measurements requirements added
  - Leveraged from ongoing OCP/Industry efforts (i.e. OCP Datacenter NVME SSD Specification)
- Addition of PLDM Type 7 / DSP0242 for File I/O as a required feature
- Minimum versions specified for referenced industry standards (PLDM, MCTP, Redfish etc.).

Additional Updates..

- AMC may act as MCTP proxy
- PLDM Type 2 – Required PDR and PLDM Event types added
- PLDM Type 5 – New firmware command requirements
- PLDM Type 6 – New RDE command requirement RDEOperationKill

Call to Action

- There is a workstream under the Hardware Management Project
  - https://github.com/opencomputeproject/OCP-HM-AMC-Auxiliary_Management_Controller
  - https://ocp-all.groups.io/g/OCP-AMC
- Contributions that would help to complement this project:
  - Additional Profiles under the HW Management Profiles workstream
  - Contribute to the smc-common code base

Thank You!

